ANALYSIS

Bùi Tiến Lên

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KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

Contents

The Big Picture

Domain situation
Observe target users using existing tools

Data/task abstraction

Visual encoding/interaction idiom
Justify design with respect to alternatives

Algorithm
Measure system time/memory
Analyze computational complexity

Analyze results qualitatively

Measure human time with lab experiment (*lab study*)

Observe target users after deployment (*field study*)

Measure adoption

Why Validate?

Validation

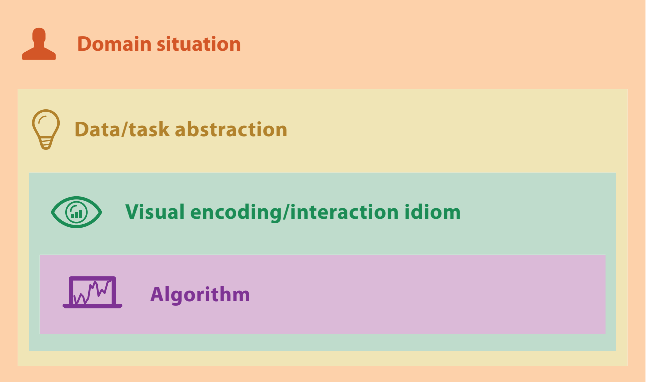
- Validation is important for the reasons: the vis design space is huge, and most designs are ineffective.
- Validation is a tricky problem that is difficult to get right.
- It's valuable to think about how you might validate your choices from the very beginning of the design process.

Four Levels of Design

- Domain Situation
- Task and Data Abstraction
- Visual Encoding and Interaction Idiom
- Algorithm

Four Levels of Design

- Splitting the complex problem of vis design into four cascading levels



Domain Situation

- The top level describe a specific domain situation, which encompasses a group of target users, their domain of interest, their questions, and their data.
- The term domain is frequently used in the vis literature to mean a particular field of interest of the target users of a vis tool, for example microbiology or high-energy physics or e-commerce.
- Each domain usually has its own vocabulary for describing its data and problems, and there is usually some existing workflow of how the data is used to solve their problems.
- The group of target users might be as narrowly defined as a handful of people working at a specific company, or as broadly defined as anybody who does scientific research.

Task and Data Abstraction

- Design at the next level requires abstracting the specific domain questions and data from the domain-specific form that they have at the top level into a generic representation.
- Abstracting into the domain-independent vocabulary allows you to realize how domain situation blocks that are described using very different language might have similar reasons why the user needs the vis tool and what data it shows.

Visual Encoding and Interaction Idiom

- At the third level, you decide on the specific way to create and manipulate the visual representation of the abstract data block that you chose at the previous level, guided by the abstract tasks that you also identified at that level.
- Each distinct possible approach is called an **idiom**.
- There are two major concerns at play with idiom design.
    - One set of design choices covers how to create a single picture of the data: the **visual encoding** idiom controls exactly what users see.
    - Another set of questions involves how to manipulate that representation dynamically: the **interaction** idiom controls how users change what they see

Algorithm

- The innermost level involves all of the design choices involved in creating an **algorithm**: a detailed procedure that allows a computer to automatically carry out a desired goal

Angles of Attack

Angles of Attack

- There are two common angles of attack for vis design: top down or bottom up.
- With **problem-driven** work, you start at the top domain situation level and work your way down through abstraction, idiom, and algorithm decisions.
- In **technique-driven** work, you work at one of the bottom two levels, idiom or algorithm design, where your goal is to invent new idioms that better support existing abstractions, or new algorithms that better support existing idioms

Threats to Validity

Threats to Validity

- Validating the effectiveness of a vis design is difficult because there are so many possible questions on the table

Validation Approaches

- Domain Validation
- Abstraction Validation
- Idiom Validation
- Algorithm Validation

Validation Approaches

Threat Wrong problem
Validate Observe and interview target users

Threat Wrong task/data abstraction

Threat Ineffective encoding/interaction idiom
Validate Justify encoding/interaction design

Threat Slow algorithm
Validate Analyze computational complexity

Implement system

Validate Measure system time/memory

Validate Qualitative/quantitative result image analysis
Test on any users, informal usability study
Validate Lab study, measure human time/errors for task

Validate Test on target users, collect anecdotal evidence of utility
Validate Field study, document human usage of deployed system

Validate Observe adoption rates

Domain Validation

- The primary threat is that the problem is mischaracterized: the target users do not in fact have these problems.
- An **immediate** form of validation is to interview and observe the target audience to verify the characterization, as opposed to relying on assumptions or conjectures
- One **downstream** form of validation is to report the rate at which the tool has been adopted by the target audience

Abstraction Validation

- The threat is that the identified task abstraction blocks and designed data abstraction blocks do not solve the characterized problems of the target audience.
- The key aspect of validation against this threat is that the system must be tested by target users doing their own work, rather than doing an abstract task specified by the designers of the vis system.

Idiom Validation

- The threat is that the chosen idioms are not effective at communicating the desired abstraction to the person using the system.
- One **immediate** validation approach is to carefully justify the design of the idiom with respect to known perceptual and cognitive principles.
- A **downstream** approach to validate against this threat is to carry out a lab study
- Another **downstream** validation approach is the presentation of and qualitative discussion of results in the form of still images or video

Algorithm Validation

- The primary threat is that the algorithm is suboptimal in terms of time or memory performance, either to a theoretical minimum or in comparison with previously proposed algorithms
- An **immediate** form of validation is to analyze the computational complexity of the algorithm, using the standard approaches from the computer science literature
- The **downstream** form of validation is to measure the wall-clock time and memory performance of the implemented algorithm

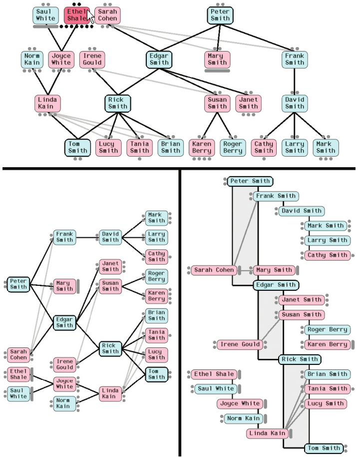
Mismatches

- A common problem in weak vis projects is a mismatch between the level at which the benefit is claimed and the validation methodologies chosen

Validation Examples

Genealogical Graphs

- McGuffin and Balakrishnan present a system for the visualization of genealogical graphs



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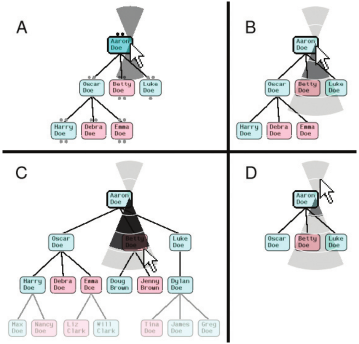


(b)

Genealogical Graphs - validation

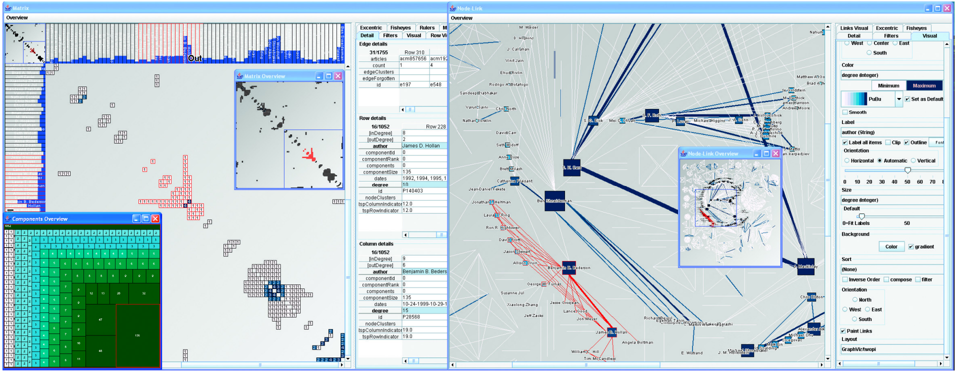
Justify encoding/interaction design

Qualitative result image analysis

Test on target users, collect anecdotal evidence of utility

MatrixExplorer

- Henry and Fekete present the MatrixExplorer system for social network analysis



MatrixExplorer - validation

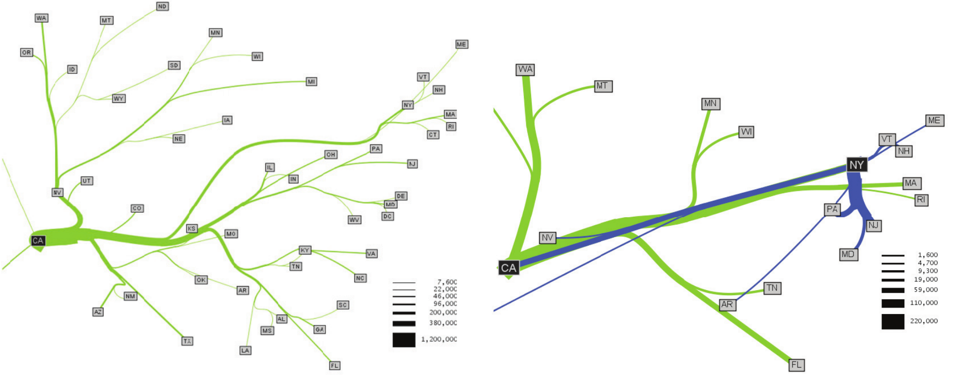
Justify encoding/interaction design

Qualitative result image analysis

Test on target users, collect anecdotal evidence of utility

Flow Maps

- Phan et al. propose a system for creating flow maps that show the movement of objects from one location to another, and demonstrate it on network traffic, census data, and trade data



(a) (b)

Flow Maps - validation

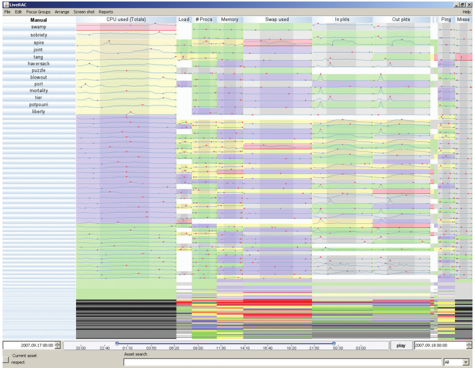
Justify encoding/interaction design

Computation complexity analysis
Measure system time/memory

Qualitative result image analysis

LiveRAC

- McLachlan et al. present the LiveRAC system for exploring system management time-series data



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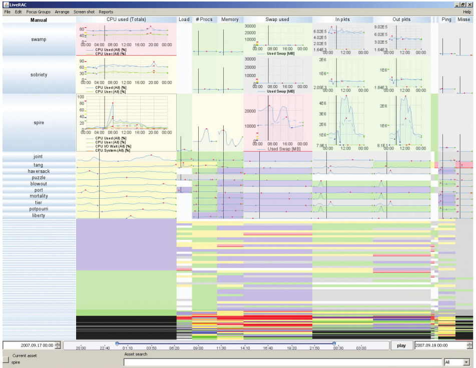


(b)

LiveRAC - validation

Observe and interview target users

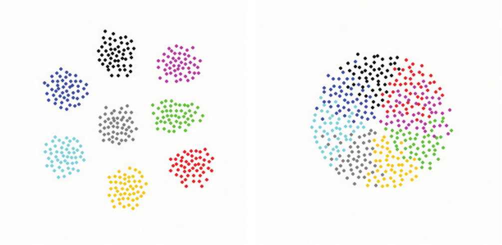
Justify encoding/interaction design

Qualitative result image analysis

Field study, document usage of deployed system

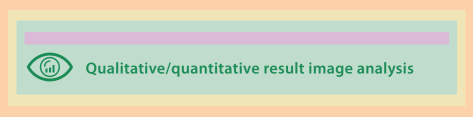
LinLog

- Noack's LinLog paper introduces an energy model for graph drawing designed to reveal clusters in the data, where clusters are defined as a set of nodes with many internal edges and few edges to nodes outside the set



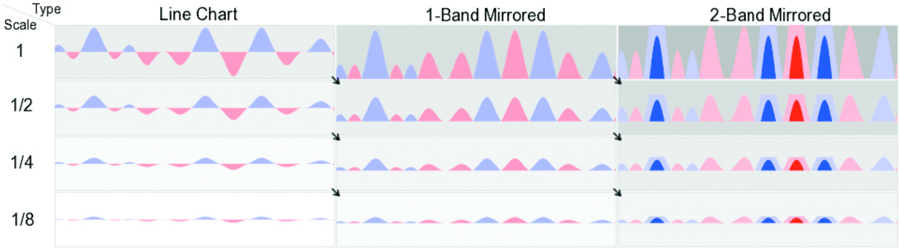
(a) (b)

LinLog - validation

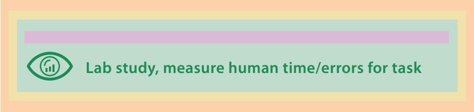


Sizing the Horizon

- Heer et al. compare line charts to the more space-efficient horizon graphs



Sizing the Horizon - validation



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